

Module 05: Action — Optimization and Control

Learning Objectives

1. Define an **objective function** and explain how minimizing or maximizing it selects actions.
2. Connect **gradient descent** to the Active Inference idea that action minimizes free energy.
3. Analyze feedback control systems using the prediction-error-correction loop.

Introduction

Action is the agent's ability to change the world. Mathematically, choosing an action is an **optimization problem**: given your beliefs about the world and your goals, which action minimizes the difference between the world as it is and the world as you want it to be? This module introduces optimization and control theory — the math behind everything from self-driving cars to the way you catch a ball.

Key Concepts

1. Objective Functions

An **objective function** (also called a cost function or loss function) assigns a number to each possible action. Lower numbers = better actions. The agent's job is to find the action that produces the minimum value.

Example: You are standing in a room with unknown temperature and you want it to be 72°F. The objective function could be: $L = (T_{\text{actual}} - 72)^2$. Any deviation from 72°F increases the cost. Turning on the heater or AC are actions that minimize L .

2. Gradient Descent

Gradient descent is the most important optimization algorithm in mathematics and machine learning. The idea is simple: measure the slope of the objective function, then take a step downhill.

$$x_{t+1} = x_t - \eta \cdot \frac{dL}{dx}$$

Where η is the **step size** (learning rate). In Active Inference, the brain performs something analogous to gradient descent on free energy: it adjusts beliefs and actions to roll downhill toward lower surprise.

3. Feedback Control (PID Controllers)

A **PID controller** (Proportional-Integral-Derivative) is the engineering cousin of Active Inference. A thermostat, cruise control, and autopilot all use this structure:

- **Proportional:** Correct based on the *current* error (prediction error)
- **Integral:** Correct based on the *accumulated* error over time
- **Derivative:** Correct based on the *rate of change* of the error

This is exactly the prediction-error-correction loop, implemented in hardware.

Applications

- **Calculus:** Finding minima and maxima of functions (set $dL/dx = 0$) is action selection: the agent is "choosing" the x that minimizes cost.
- **Machine Learning:** Training a neural network is gradient descent on a loss function. Each training step is an "action" that adjusts the weights.

Discussion Questions

1. A cruise control system is set to 65 mph but the car is going 55 mph up a hill. Describe each step of the PID correction.
2. What goes wrong if the learning rate η is too large? Too small?

Summary

Action is optimization: choosing the behavior that minimizes an objective function. Gradient descent is the algorithm that implements this. Feedback control systems (thermostats, cruise control, Active Inference agents) all share the same prediction-error-correction loop.

References

- Boyd, S. & Vandenberghe, L. (2004). *Convex Optimization*. Chapter 1.